

Systems Thinking Principles

Principle	Description	Prompting Questions
Interconnectedness	All parts of a system are connected and affect each other	• How does a change in one area affect others?• What happens if we change X? What else might change as a result?
Feedback Loops	Circular processes where outputs affect inputs	• Where do you see reinforcing loops (problems making other problems worse)?• Are there any balancing loops that keep things stable?
Leverage Points	Places in the system where small changes create big effects	• Where could a small change make the biggest difference?• Which part seems to influence many other parts?
Unintended Consequences	Changes can create unexpected effects elsewhere	• What might be some unexpected results of your solution?• Who might be affected that we haven't considered?
Boundaries	Systems exist within larger systems	• What factors outside the system affect this system?• How does this system connect to other systems?
Time Delays	Changes don't always show immediate effects	• How might impacts differ in the short-term versus long-term?• Which problems might take time to emerge?
Multiple Perspectives	Different stakeholders see the system differently	• How does this look from different viewpoints (students, staff, administrators)?• What priorities might different groups have?

Explaining Feedback Loops in Systems Thinking

Feedback loops are circular patterns of cause and effect where the output of a process affects the input to that same process. There are two main types:

1. Reinforcing Loops (Snowball Effect)

Definition: A reinforcing loop amplifies change in one direction. Changes in one component lead to more changes in the same direction, creating a cycle that can be either virtuous (positive) or vicious (negative).

2. Balancing Loops (Stabilizing Effect)

Definition: A balancing loop counteracts change to maintain equilibrium. When something increases, other factors work to bring it back down, or vice versa.